INNOVATIVE ASSIGNMENT REPORT

3CS103ME24 – Data Analysis and Visualisation

SteelScope

Visual Analytics of 1,389 Steels and Iron Alloys

Python notebooks · KNIME · Tableau · Power BI · Web application

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Abstract

This project analyses a public table of steels and iron alloys (2,672 rows, 34 columns: grade designation, the weight-percent of 30 chemical elements, tensile strength, liquidus temperature and a strength class) with the aim of answering three engineering questions: which elements drive strength, which grades can substitute for one another, and which families can deliver a required strength and melting range. A logged cleaning pipeline found that the data hid three problems: 62 exact duplicate rows, 1,221 grades listed twice under two spellings of the same name, and 469 strength values (18 %) that were the column mean copied into empty cells. After cleaning, the analysis works on **1,389 canonical grades, 1,382 of them with a measured strength**.

The cleaned data is explored in three Python notebooks with 39 figures covering every technique of the course: the five-number summary, binning, chi-square tests with Cramér's V, correlation and linear regression, principal component analysis (PCA), and radar, chord, ring, treemap, ridgeline and waterfall charts. The same statistics are rebuilt as a KNIME workflow, which also holds all machine learning (k-means clustering, decision tree, k-nearest neighbours and random forest on a group-safe train/test split). Interactive dashboards were built in Tableau and Power BI, and a web application (SteelScope) lets any user explore the grades, inspect one grade, find chemically similar substitutes and search for grades that meet a target.

The main findings are: the **family decides the strength range** (median strength from about 260 MPa for grey cast iron to about 1,790 MPa for spring steel); **carbon is the strength dial of plain steels** (+43 MPa per 0.1 % carbon, $R^2 = 0.51$) although pooled across all steels it shows no correlation (a Simpson's paradox); the family–strength link is strong ($\chi^2 = 548.7$, Cramér's V = 0.45); and composition alone predicts strength with $R^2 = 0.73$ (random forest) while the remainder is due to heat treatment, which the data does not record.

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1. Introduction

1.1 Motivation

Engineers and buyers choose steel from thousands of standard grade names (EN, SAE-AISI, ASTM, UNS, AWS, ACI). A grade name does not show its chemistry or strength, two names can describe the same steel, and the same chemistry can appear under several names. A data-driven view of many grades at once makes these choices faster and more transparent. The course asks for a visualization-intensive analysis carried across several tools; this project applies it to a real engineering dataset.

1.2 Objectives

Table 1: *The three questions the project answers*

#	Question	Where it is answered
1	**Drivers:** which elements make a steel strong, and what controls its melting point?	Notebook 02 (Q1–Q5), notebook 03 (Acts IV–V), KNIME sections E–F
2	**Substitution:** which grades have a similar chemistry, so one can replace another?	PCA map (notebook 03), k-means and k-NN in KNIME, web app substitute finder
3	**Target design:** for a required strength and melting range, which families deliver it?	Notebook 02 (Q5), Power BI target finder, web app target designer

1.3 Tools and division of work

Table 2: *Tools used and their role*

Tool	Role in the project
Python (pandas, NumPy, SciPy, scikit-learn, matplotlib, seaborn)	Cleaning, statistics and 39 figures in three notebooks; exports for the other tools
KNIME Analytics Platform 5	The statistics rebuilt as a node workflow, plus all machine learning
Tableau	Three interactive workbooks (dashboards) on the exported data
Power BI	A six-page interactive report on a star-schema model
Web application (Next.js, React, shadcn/ui, Recharts)	SteelScope: explorer, grade inspector, substitute finder, target designer

Data flows in one direction: raw data → notebook 01 (cleaning) → processed data → notebooks 02 and 03 → exported tables → KNIME, Tableau, Power BI and the web application. Nothing downstream is edited by hand; a correction is made in the notebook and the chain is re-run.

2. Domain background

2.1 Steel and cast iron

Steel is iron alloyed with a small amount of carbon (typically up to about 2.1 wt%) and other elements. Carbon atoms sit between iron atoms and hinder the movement of crystal defects, which raises strength and hardness. Above about 2.1 % carbon the alloy is a cast iron: the excess carbon forms graphite (flakes in grey iron, nodules in ductile iron), which makes casting easy but usually lowers strength and ductility.

Properties depend not only on chemistry but on the microstructure produced by heat treatment: soft ferrite, tough austenite (stable at room temperature when enough nickel is present, as in 304 and 316 stainless steels), pearlite, and hard martensite formed by rapid quenching. Processes such as quenching, tempering, ageing and precipitation hardening change strength without changing composition. The dataset records composition but not heat treatment; this sets a limit on what any composition-based model can achieve.

2.2 Role of the alloying elements

Table 3: *Effect of the main alloying elements*

Element	Main effect
C (carbon)	Main strengthener of plain steels; raises hardenability and lowers weldability
Mn (manganese)	Deoxidiser; adds strength and hardenability
Si (silicon)	Deoxidiser; promotes graphite in cast irons
Cr (chromium)	From about 10.5 % forms a protective oxide film (stainless steel); raises hardenability
Ni (nickel)	Stabilises austenite; improves toughness and corrosion resistance
Mo (molybdenum)	Hardenability, high-temperature strength, pitting resistance
V, W (vanadium, tungsten)	Form hard carbides; tool and high-speed steels keep hardness when hot
Co (cobalt)	Hot hardness in high-speed steels; strengthening in maraging steels
N (nitrogen)	Austenite stabiliser and strengthener in stainless steels
Cu, B, Nb, Ti	Weathering resistance (Cu); hardenability at ppm level (B); grain refinement (Nb, Ti)

2.3 The six family groups

The 17 families in the data are combined into six groups used in every tool (the same rule in Python, the KNIME Rule Engine, Tableau, Power BI and the web application):

Table 4: *Family groups (all 1,389 grades)*

Group	Families	Grades
Carbon & low-alloy	Carbon/non-alloy, alloy, boron, bearing, spring, nitriding, weathering	583
Stainless	Stainless steel (austenitic, ferritic, martensitic, duplex, precipitation hardening)	546

Group	Families	Grades
Cast iron	Grey, ductile, malleable and austenitic (Ni-Resist) cast iron	67
Tool & high-speed	Tool steel, high-speed steel	82
Weld metal	AWS filler and electrode deposits	83
Other	Maraging steel, Fe-Ni/Co special alloys	28

2.4 Engineered indices

Carbon equivalent (CE). The International Institute of Welding (IIW) formula combines the elements that raise hardenability into one number: **CE = C + Mn/6 + (Cr + Mo + V)/5 + (Ni + Cu)/15**. A high CE means the heat-affected zone of a weld can turn into brittle martensite and crack. Bands used: CE ≤ 0.40 readily weldable, 0.40–0.60 preheat advised, above 0.60 difficult. The bands are meaningful only for carbon and low-alloy steels; in stainless steels chromium is present for corrosion resistance and the CE overstates hardenability.

Chromium and nickel equivalents (Schaeffler). Cr_eq = Cr + Mo + 1.5·Si + 0.5·Nb sums the ferrite formers and Ni_eq = Ni + 30·C + 0.5·Mn the austenite formers; their balance predicts whether a stainless steel is austenitic, ferritic, martensitic or duplex. **Alloying total** = 100 − Fe, the weight-percent of everything that is not iron.

3. Dataset

The source file Alloys.csv has **2,672 rows and 34 columns**: the grade designation, the weight-percent of 30 elements (Al, As, B, C, Ca, Ce, Co, Cr, Cu, Fe, La, Mg, Mn, Mo, N, Nb, Ni, O, P, Pb, S, Se, Si, Sn, Ta, Ti, V, W, Zn, Zr), the tensile strength in MPa, the liquidus temperature in °C and a three-level strength class. The schema check found no missing values and no negative percentages; iron ranges from 29.3 % to 99.97 %.

At first sight the table is complete, but three problems were hidden in it (Section 4): duplicate rows, the same grade listed twice under two spellings of its name, and strength values that had been filled in with the column average. A fourth issue concerns the liquidus: 1460 °C is used for 524 grades as a placeholder rather than a measured value.

Table 5: *Key variables*

Term	Meaning
Tensile strength (TS)	Maximum stress before the material breaks in a tension test, in MPa (1 MPa = 1 N/mm²)
Liquidus	Temperature above which the alloy is completely liquid
Strength class	Rebuilt from measured TS: Low < 300 MPa, Medium 300–600 MPa, High > 600 MPa
wt%	Weight percent of an element in the alloy

4. Data cleaning pipeline

Cleaning is implemented in notebook 01 as a sequence of small steps; every step records the number of rows before and after and a one-line explanation in a cleaning log (reports/cleaning_log.json).

Table 6: *Cleaning steps and their effect*

Step	Finding	Action	Rows after
Load and validate	34 columns, numeric, no nulls, no negative wt%	Accepted	2,672
Exact duplicates	62 identical rows	Removed	2,610
Imputed strength	469 rows (18 %) have TS = 652.482978, exactly the raw column mean	Flagged as not measured	2,610
Twin listings	1,221 grades appear twice (name with spaces and with hyphens), 168 once	Canonical grade key; merged	1,389
Recover strength	459 grades had an imputed listing whose twin carries the measured value	Measured value kept	1,389
No measured strength	7 grades have no measured TS at all	Kept, excluded from strength statistics	1,382 measured
Strength class	Class rebuilt from measured TS (edges 300 and 600 MPa); agrees 100 % with the original on measured listings	Counts Low/Medium/High = 23 / 648 / 711	–
Liquidus 1460 °C	Does not fall with carbon as a measured liquidus should; tied to families and name formats	Flagged as placeholder (524 grades)	–
Features	–	Alloying total, CE, Cr_eq, Ni_eq added	–

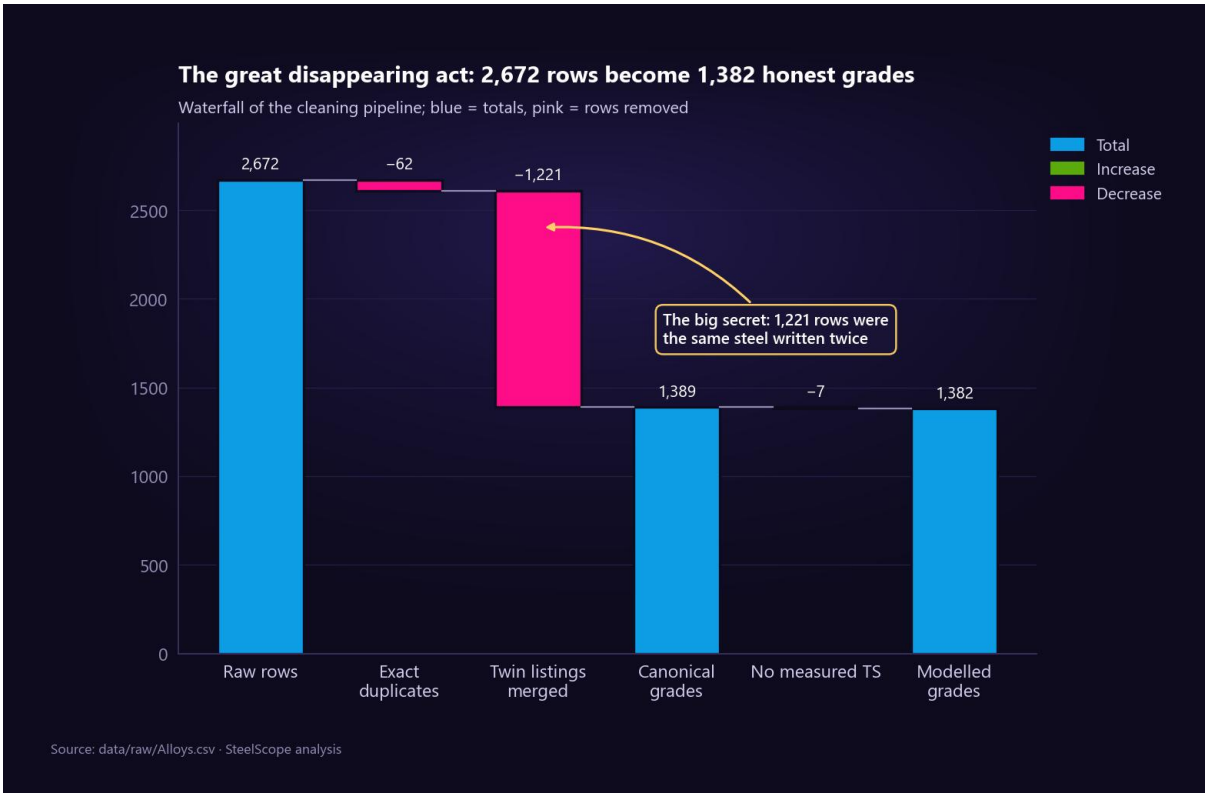


Figure 1: *Cleaning waterfall: 2,672 raw rows shrink to 1,382 grades with a measured strength; the twin listings are the largest drop.*

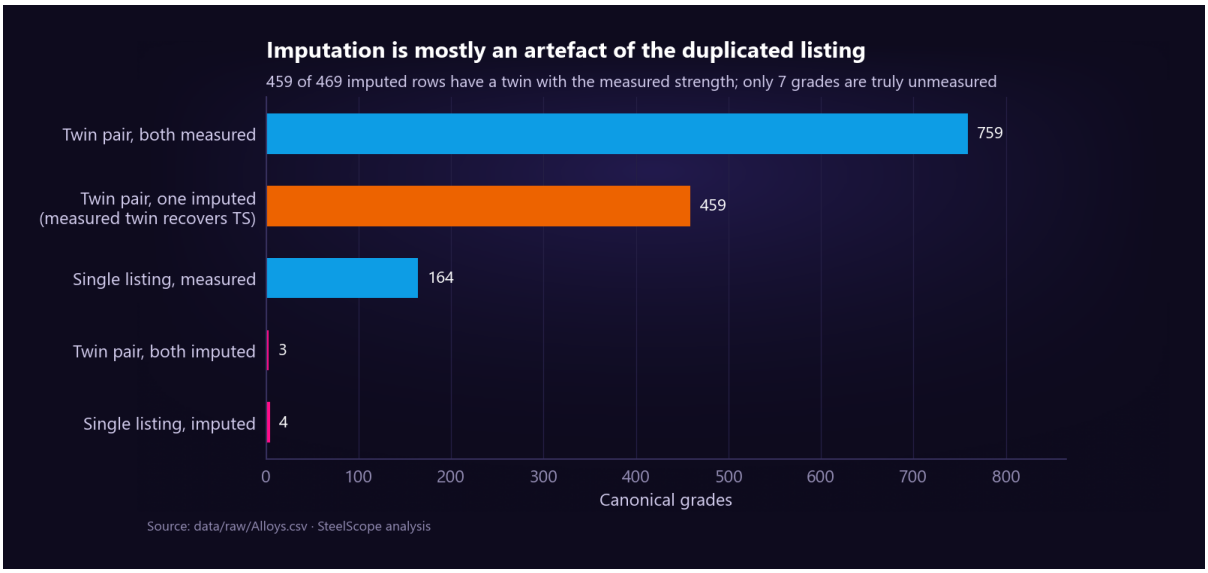


Figure 2: *Twin listings and the imputed value: the fake 652 MPa values sit mostly in one listing of a twin pair.*

Irreducible error. Some compositions occur with several strengths: maraging steel aged and annealed, the CB7Cu cast stainless in its H900…H1150 tempers, cast irons poured from one nominal chemistry. Their strength depends on processing, which is not in the data, so no composition-based model can separate them. After removing the imputed constant, 28 composition groups still carry different measured strengths at grade level. These grades define the ceiling of every strength model built later.

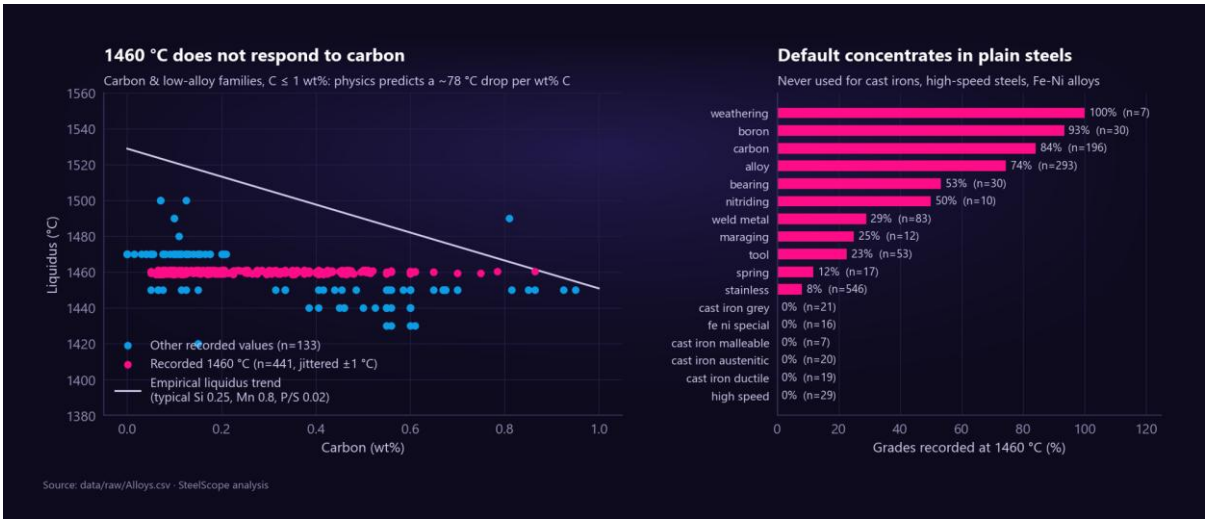


Figure 3: *The 1460 °C liquidus: a placeholder rather than a measurement (it does not follow carbon content).*

5. Exploratory data analysis

Notebook 02 asks seven questions an engineer would ask; each is answered with one statistic and one figure rather than many unfocused charts.

Table 7: *The seven exploratory questions*

Q	Question	Answer
Q1	How much does family alone decide strength?	Family is the first-order factor: family medians span about sevenfold, from grey cast iron to spring steel
Q2	Which elements drive strength? Does pooling mislead?	Pooled, carbon looks irrelevant (Spearman ρ = −0.03); inside plain steels it is the main driver (ρ = 0.43): Simpson's paradox
Q3	Stainless: does more alloy mean more strength?	No; the microstructure set by the Cr–Ni balance matters. High-nickel austenitic grades are not the strong option
Q4	Is the carbon equivalent a strength index?	A reasonable proxy for plain and tool steels; meaningless for stainless and cast irons
Q5	Strength versus liquidus	Nearly independent design axes; liquidus mostly separates families
Q6	What separates the strength classes?	Family, not a single element; the low class is almost entirely cast iron
Q7	What if the imputed rows were kept?	A fake peak at 652 MPa, family medians equal to the imputed value, weaker element correlations

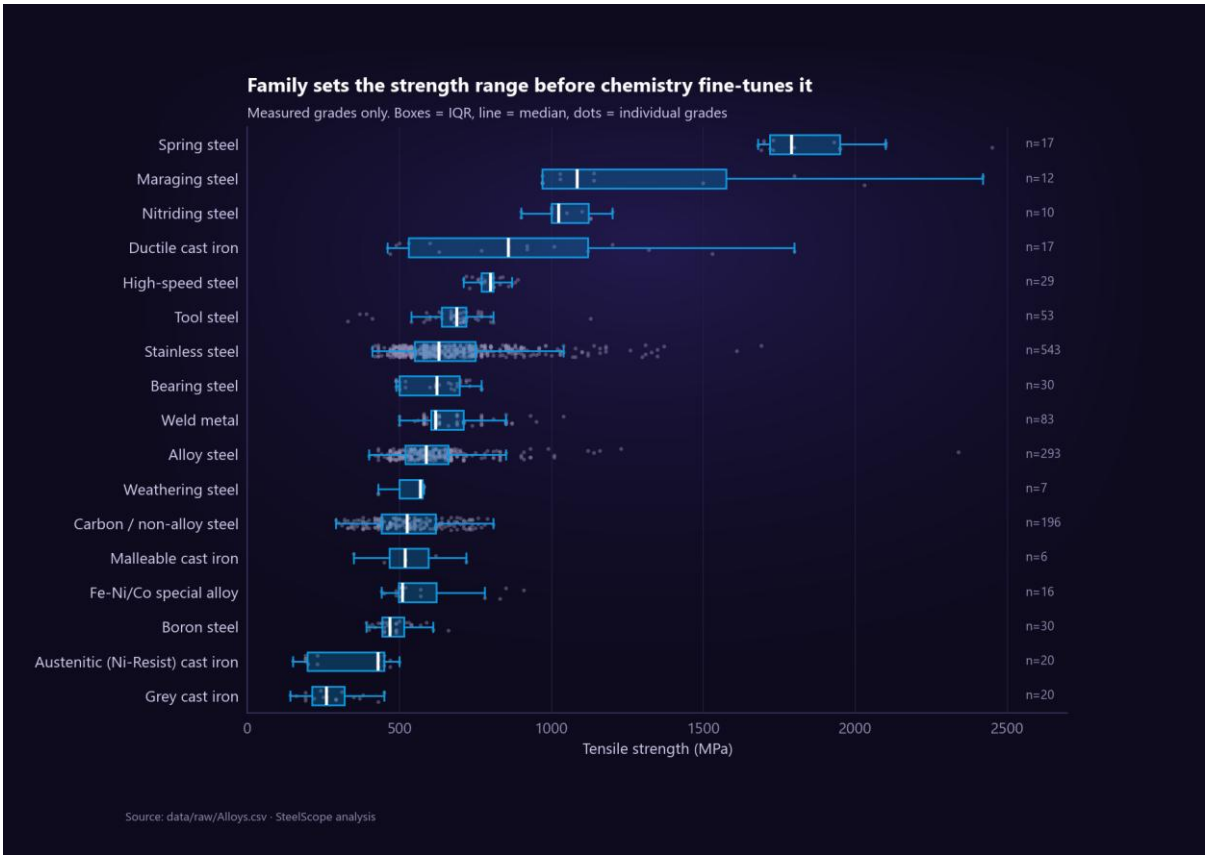


Figure 4: *Q1: tensile strength by family; the family sets the strength range.*

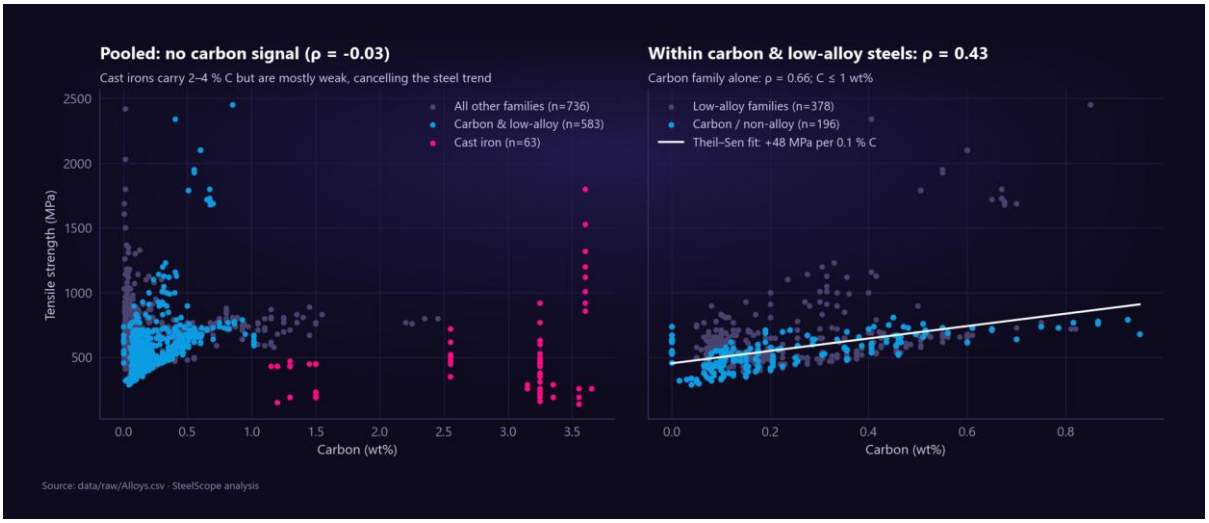


Figure 5: *Q2: Simpson's paradox. Pooled over all steels carbon shows no trend; within carbon and low-alloy steels it is the main strengthener.*

Why the paradox happens. Cast irons contain 2–4 % carbon (as graphite) yet are mostly weak, while plain steels gain strength with carbon. Mixing the two groups cancels the trend. Every element–strength statement therefore has to be made within a family, and every strength model needs the family as an input.

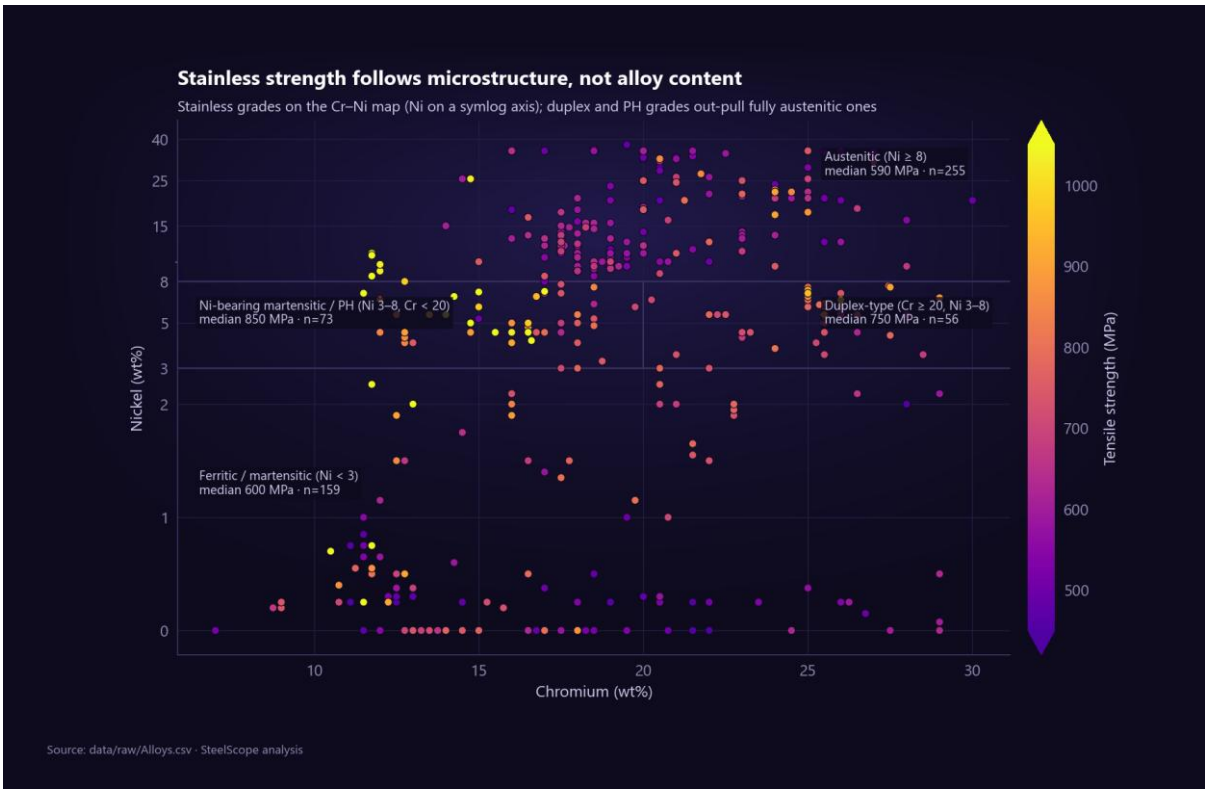


Figure 6: *Q3: chromium and nickel equivalents of stainless grades. Strength follows the microstructure, not the amount of alloy.*

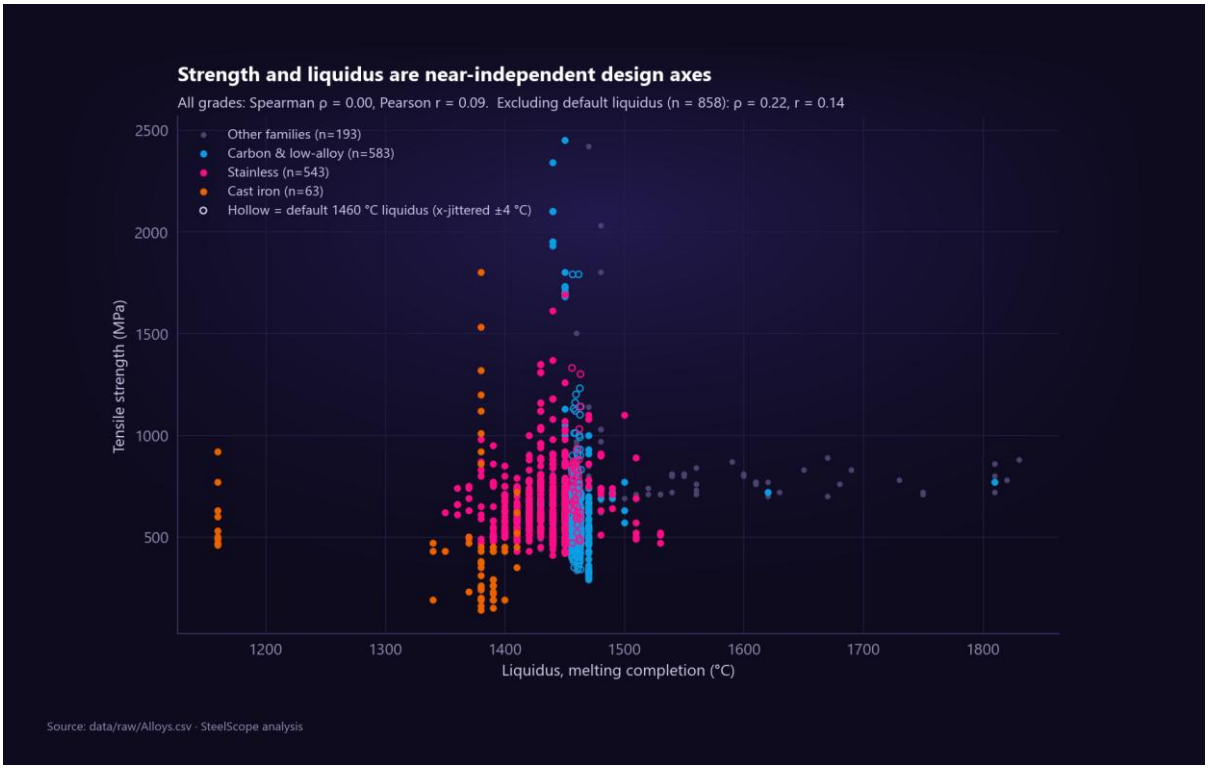


Figure 7: *Q5: strength versus liquidus; the two are nearly independent, so a target search must filter on both.*

6. Statistical analysis and visual story

Notebook 03, "The Secret Lives of 1,389 Steels", tells the analysis as a story in a prologue, seven acts and an epilogue with 27 figures, and finishes by exporting the tables used by Tableau, KNIME and Power BI.

6.1 Descriptive statistics

Table 8: *Five-number summary and moments of tensile strength*

Statistic	Tensile strength (1,382 grades)
Minimum / Q1 / median / Q3 / maximum	140 / 510 / 610 / 720 / 2,450 MPa
Mean / standard deviation	651.6 MPa / 242.5 MPa
Skewness / kurtosis	2.89 / 13.7 (long right tail, many extreme grades)
Outliers (1.5 × IQR rule)	75: 65 above 1,035 MPa (spring, maraging, high-strength alloy steels), 10 below 195 MPa (weakest cast irons)

Because the distribution is strongly right-skewed (mean above median, skewness 2.89), the project reports medians and interquartile ranges rather than means and standard deviations. The outliers are real steels and are kept.

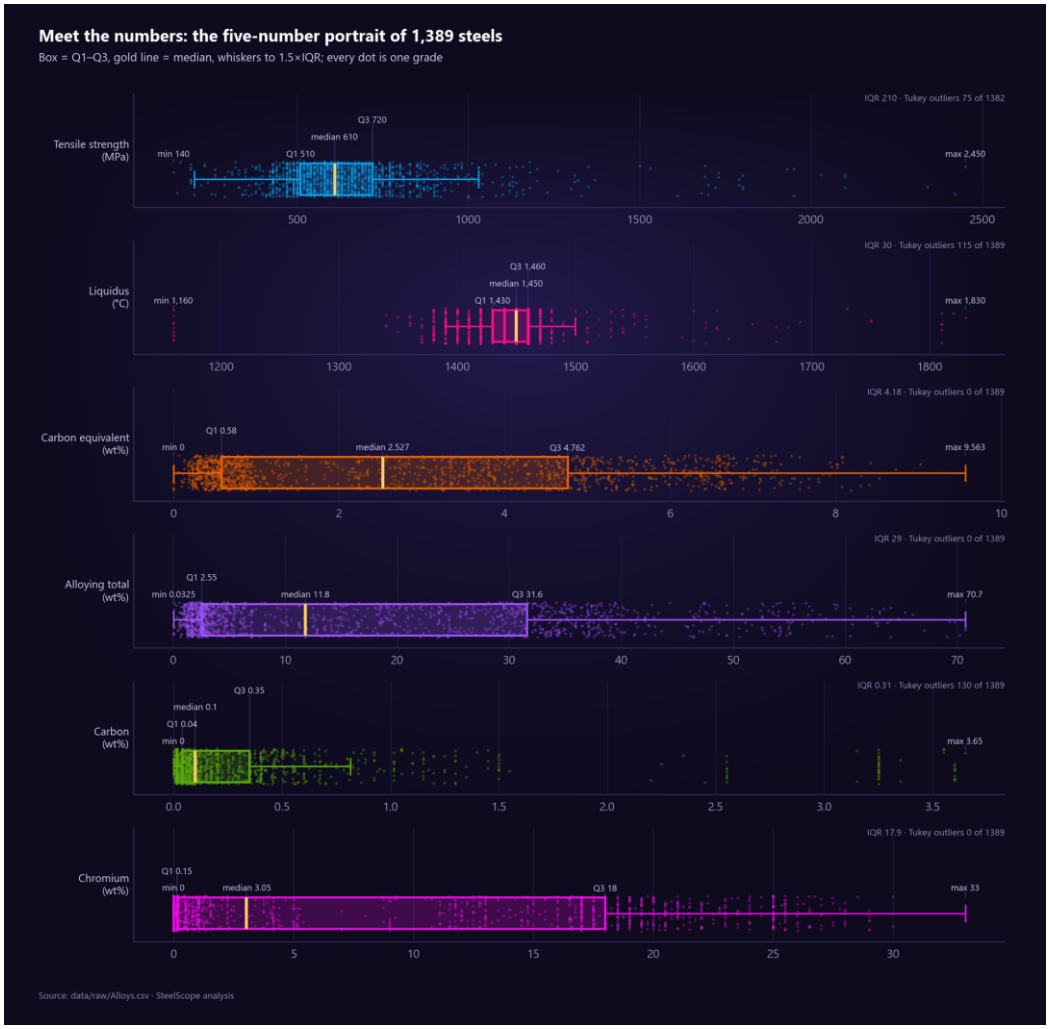


Figure 8: *Five-number summaries of the main variables.*

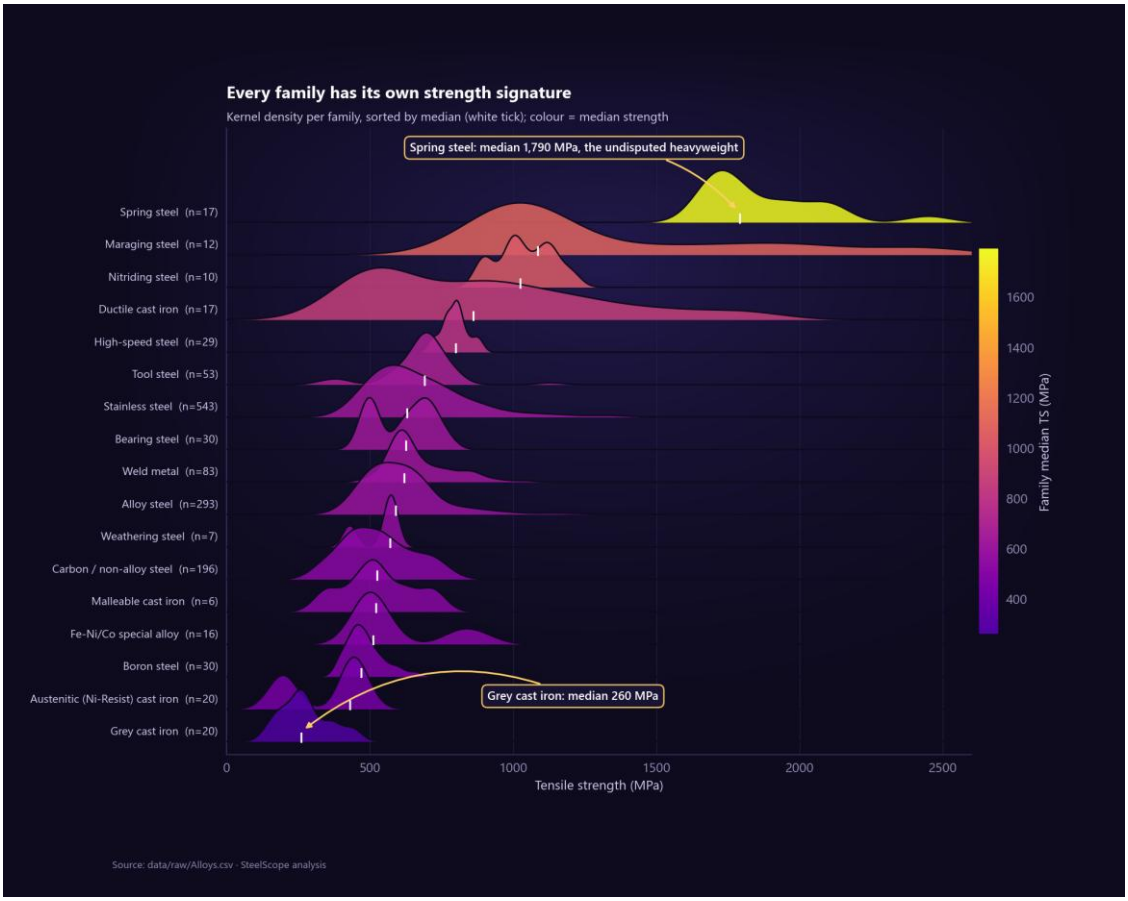


Figure 9: *Ridgeline of strength per family: every family has its own strength signature.*

6.2 Binning

Three binning strategies were compared on tensile strength. **Equal width** (5 bins of 462 MPa) puts 96 % of the grades into the first two bins (671 / 650 / 36 / 18 / 7) because of the skew. **Equal frequency** (quantile edges at 500, 580, 650 and 750 MPa) gives 318 / 284 / 240 / 268 / 272; the counts are not identical because many grades share round values such as 500 or 580 MPa. **Domain bins** carry metallurgical meaning: carbon bands (low ≤ 0.25 %, medium 0.25–0.60 %, high 0.60–2.1 %, cast-iron range > 2.1 %) with 967 / 254 / 112 / 49 grades, and the CE weldability bands (173 / 163 / 247 among the 583 plain steels).

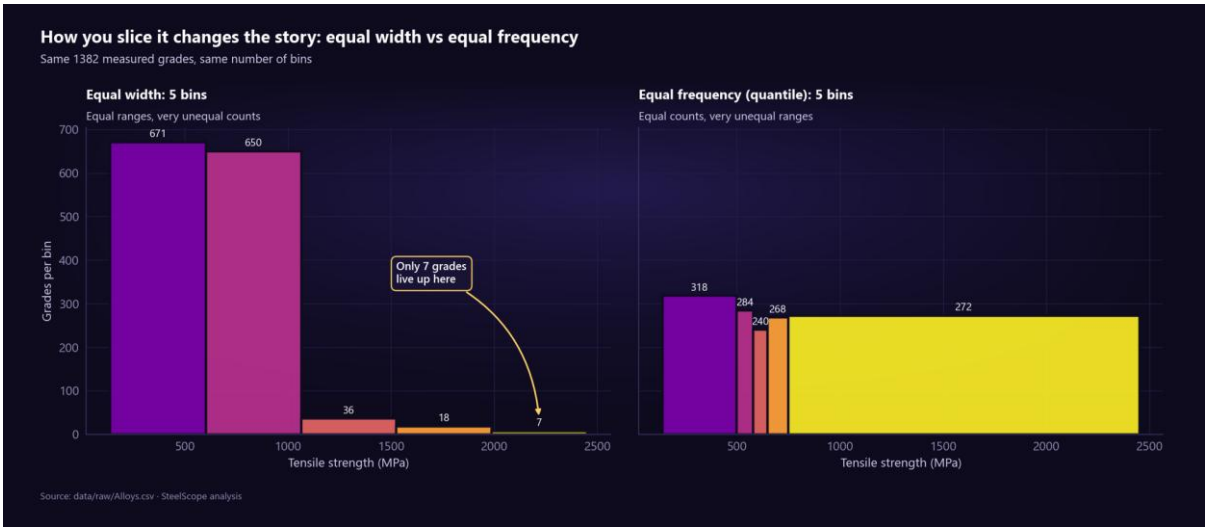


Figure 10: *Equal-width versus equal-frequency binning of tensile strength.*

6.3 Chi-square tests of independence

For each pair of categorical variables the null hypothesis is that they are unrelated. The chi-square statistic compares observed counts with the counts expected under independence (row total × column total ÷ n); Cramér's V rescales it to 0–1 as a measure of association strength. Because the low-strength class has only 23 grades, four tables have more than 20 % of expected counts below 5 (Cochran's rule), so each result was confirmed with a Monte-Carlo permutation test (labels shuffled thousands of times); chance never produced an association as strong.

Table 9: *Chi-square tests*

Pair	χ^2	df	p-value	Cramér's V	n
Family group × strength class	548.73	10	1.7×10^{-111}	0.45	1,382
Product form × strength class	117.37	4	2.0×10^{-24}	0.21	1,382
Carbon band × strength class	315.85	6	3.3×10^{-65}	0.34	1,382
CE band × strength class (plain steels)	198.22	4	9.1×10^{-42}	0.41	583
Family group × default liquidus	631.97	5	2.5×10^{-134}	0.67	1,389

All five null hypotheses are rejected. The largest standardized residual is cast iron × low class: 22 grades where independence predicts about one (residual +20.5). The strongest association is family × default liquidus (V = 0.67), confirming that the 1460 °C value is a family habit, not a measurement.

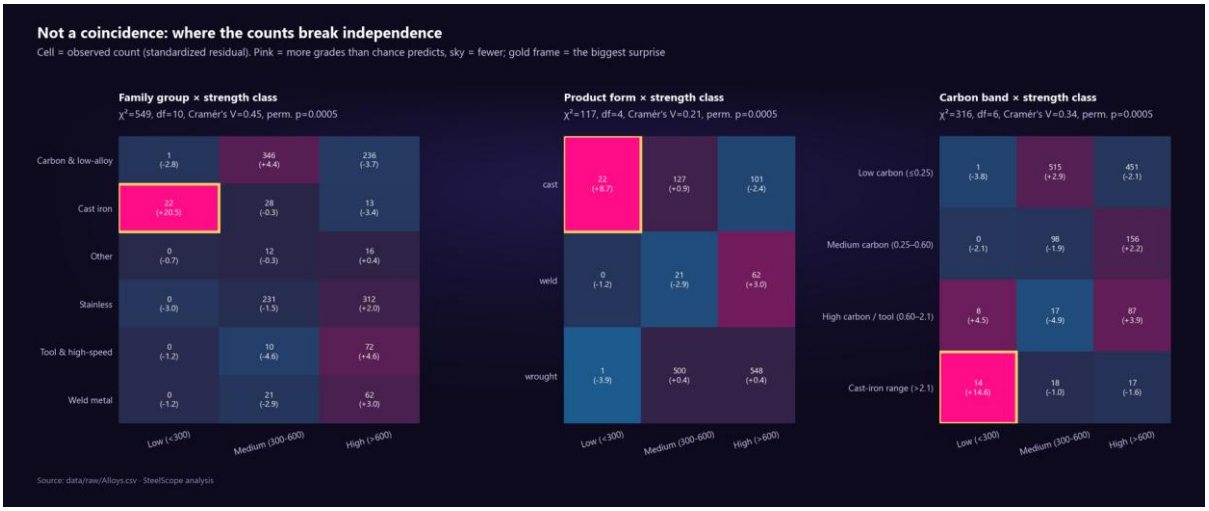


Figure 11: *Standardized residuals of the chi-square tests; red cells hold more grades than independence predicts.*

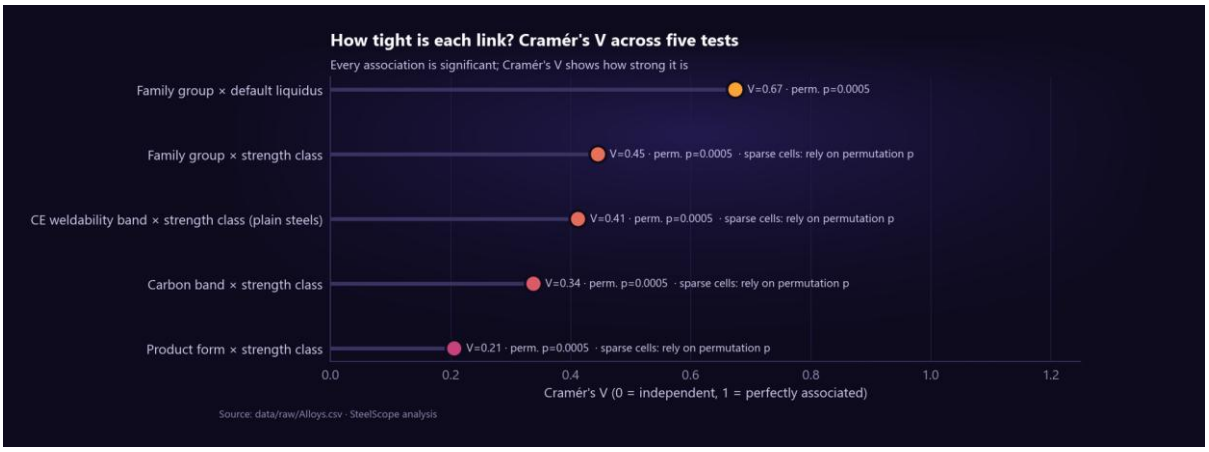


Figure 12: *Cramér's V ranks the strength of each association.*

6.4 Correlation and regression

Pearson correlations between strength and any single element are weak across all 1,382 grades (all below 0.19 in absolute value; carbon −0.02), while strong blocks appear among the elements themselves: carbon equivalent with alloying total (0.94), chromium with carbon equivalent (0.92), and the vanadium–tungsten pair of tool steels (0.55).

Table 10: *Linear regression results*

Model	Data	Result
Simple: TS ~ C	Carbon steels (n = 196)	TS = 429.2 + 428.6 × C; R^2 = 0.508; p(C) ≈ 10^{-31} → about +43 MPa per 0.1 % carbon
Multiple: TS ~ C, Mn, Si, Cr, Ni, Mo, V	Carbon & low-alloy, C ≤ 1 % (n = 574)	Coefficients (MPa per wt%): intercept 404.0, C 648.5, Mn −19.6, Si 193.3, Cr 19.7, Ni 23.8, Mo −74.2, V 362.2; R^2 = 0.326, adjusted R^2 = 0.317

Carbon alone explains about half of the strength variance in carbon steels; seven elements together explain only a third across plain steels, because heat treatment is missing from the data. This motivates the non-linear models built in KNIME.

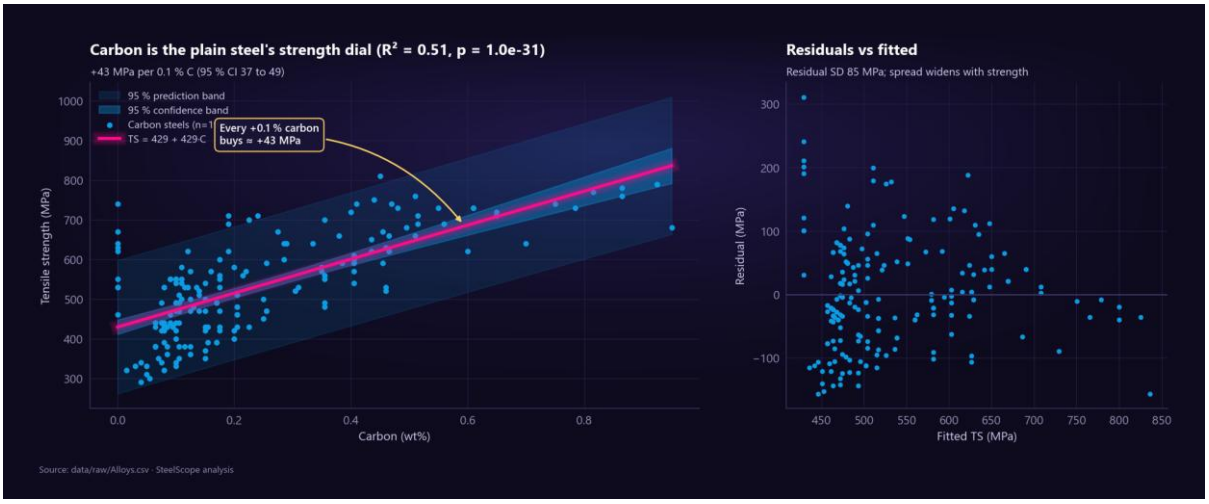


Figure 13: *Simple regression of strength on carbon in carbon steels, with confidence and prediction bands.*

6.5 Principal component analysis

The 20 non-rare composition features (all elements except iron and ten rare traces, plus their summed trace total) were transformed with log(1 + x), because a few grades hold very large amounts of chromium, tungsten or nickel, and standardised to z-scores so that each element has equal weight. PCA then finds new axes of maximum variance.

Table 11: *Variance explained by the first five components*

Component	PC1	PC2	PC3	PC4	PC5
Variance explained	15.0 %	11.8 %	9.5 %	7.6 %	5.9 %

Twelve components are needed to reach 80 % of the variance and eight have an eigenvalue above 1 (Kaiser rule): steel chemistry is genuinely high-dimensional. PC1 is driven by chromium (+0.80), nickel (+0.77) and nitrogen (+0.69) against carbon (−0.58), so stainless steels stretch along it; tool and high-speed steels rise along PC2. Differently named plain steels crowd together in the centre of the map: **names are not chemistry**.



Figure 14: *Scree plot and loadings of the principal components.*

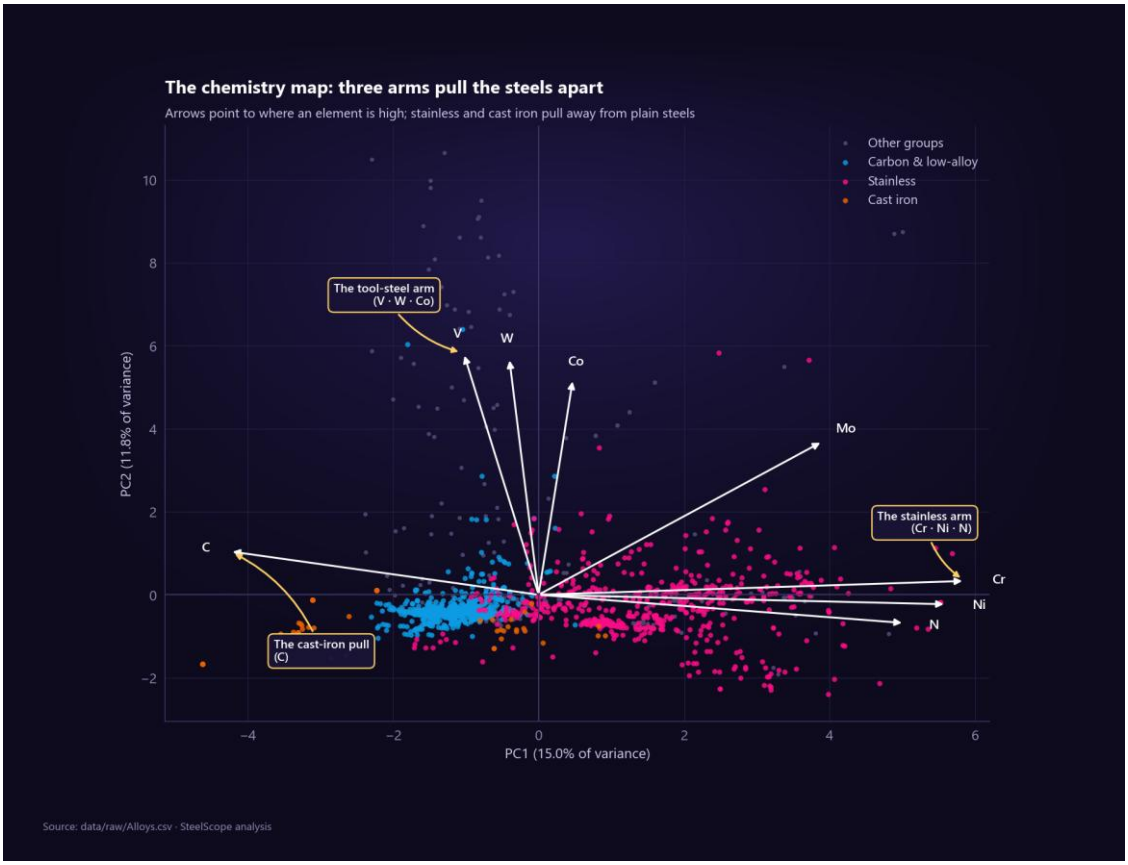


Figure 15: *The chemistry map: PCA biplot of 1,389 grades coloured by family group.*

6.6 Composition portraits and circular charts

Radar charts give each alloy or group a composition fingerprint. Nine showcase grades (AISI 1018, 1045, 4140, 304, 316, 17-4 PH, M2 high-speed steel, grey iron 250 and aged maraging 300) are compared on strength, liquidus, CE, alloying total and key elements. The stainless overlay shows that 17-4 PH reaches a higher strength than nickel-rich 304 and 316 with less nickel: **strength is not bought by the kilogram of alloy**. The treemap and chord diagram summarise the population and the effect of cleaning.

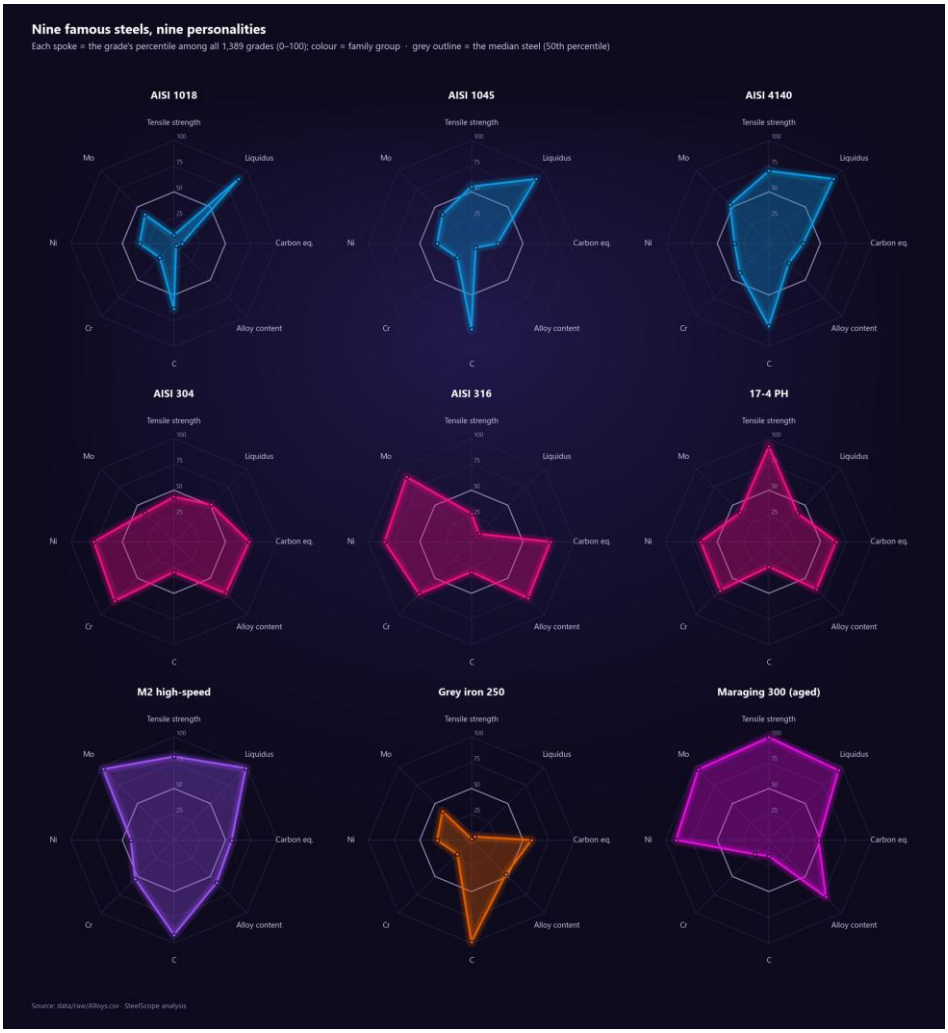


Figure 16: *Radar portraits of nine showcase alloys.*

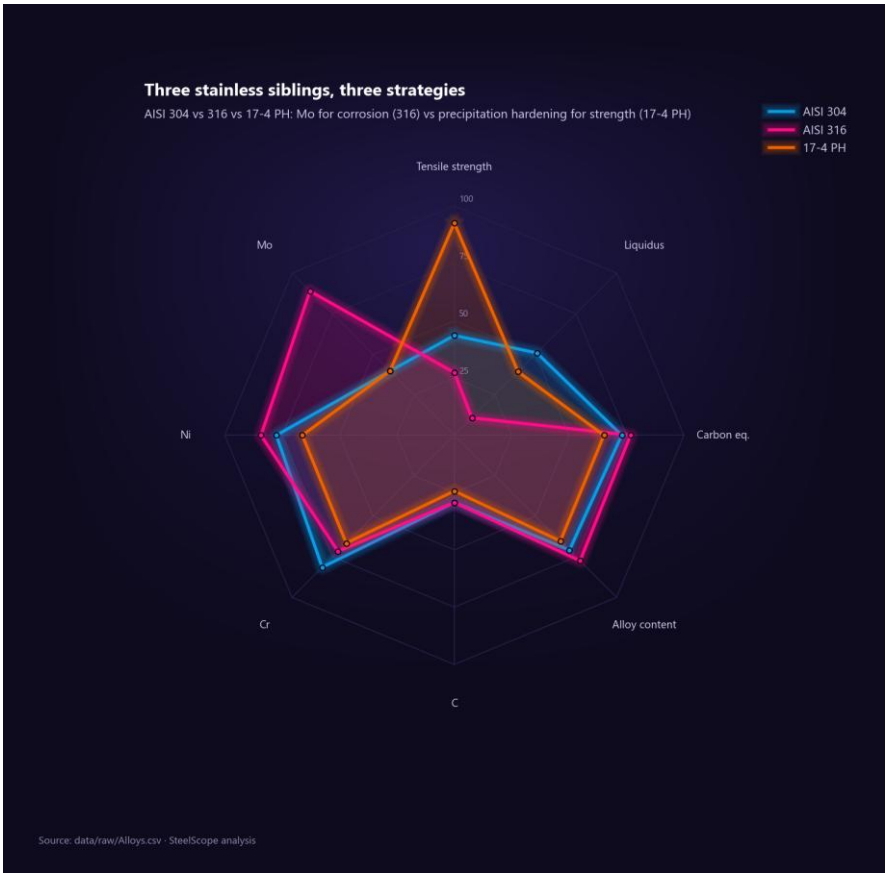


Figure 17: *Stainless overlay: 17-4 PH versus 304 and 316.*



Figure 18: *Treemap of families: tile size = number of grades, colour = median strength.*

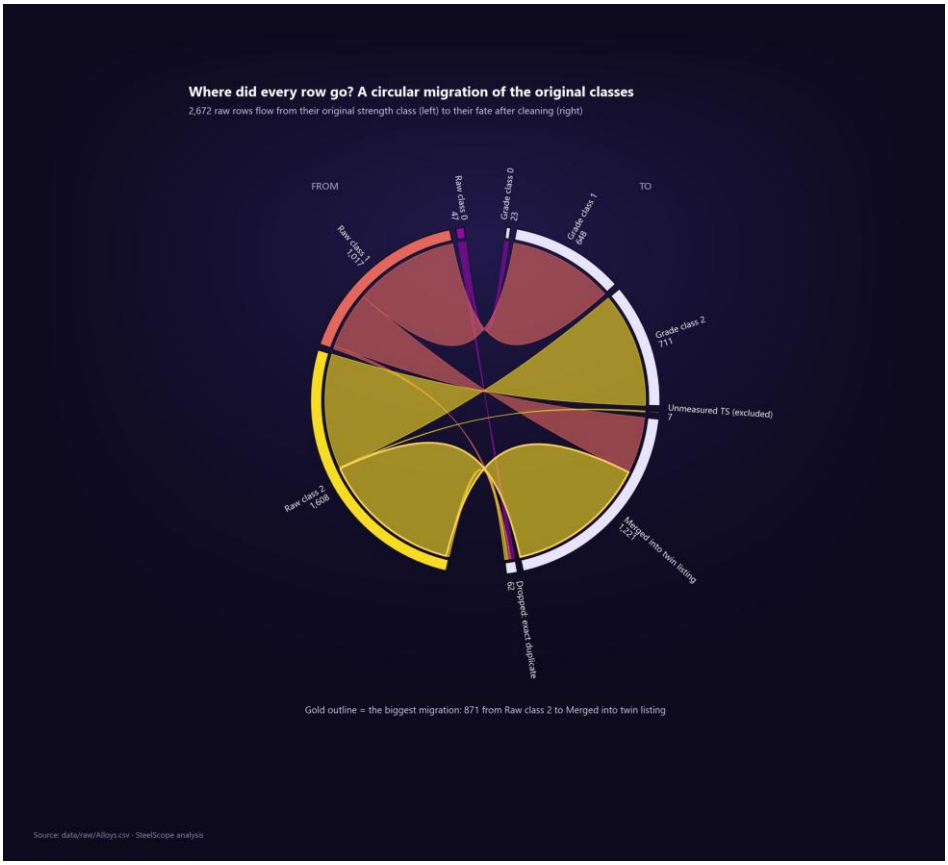


Figure 19: *Chord diagram of how the original strength-class labels move after cleaning.*

6.7 The carbon-equivalent waterfall

For SAE-AISI 4140 (a Cr-Mo steel) the waterfall adds the CE terms one by one: carbon 0.405 + Mn/6 0.146 + Cr/5 0.190 + Mo/5 0.040 = **0.78**, in the "difficult" band. Carbon and chromium contribute most, so a welder must preheat.



Figure 20: *Carbon-equivalent waterfall for AISI 4140.*

7. KNIME workflow

The KNIME workflow (knime/workflow/SteelScope_Visual_Analytics.knwf) rebuilds the statistics as nodes and holds all machine learning. It reads knime_input.csv (1,389 grades with a group-safe train/test split column) from its own workflow data area, so it runs unchanged on another computer.

Table 12: *Sections of the KNIME workflow*

Section	Nodes	Result
A · Read, clean, enrich	CSV Reader, Row Filter, Rule Engine, Math Formula ×5, Color Palette Designer, Color Designer (Apply)	1,382 measured grades; six family groups; KNIME's CE equals Python's exactly (maximum difference 0)
B · Five-number summary	GroupBy, Statistics View, Box Plot, Histogram, Numeric Outliers	140 / 510 / 610 / 720 / 2,450 MPa; 75 outliers
C · Binning	Binner (equal width, custom quantiles), Numeric Binner, Bar Chart	671/650/36/18/7 and 318/284/240/268/272 ("to lower bin" tie rule)
D · Chi-square	Crosstab, Python View residual heatmap	$\chi^2 = 548.7$, df = 10
E · Correlation, regression	Linear Correlation, Linear Regression Learner, Regression Predictor, Scatter and Line Plot	Slope 428.6 MPa per wt% C, R^2 0.51
F · PCA	Math Formula (Multi Column) log, Normalizer z-score, PCA Compute/Apply, scree Bar Chart	PC1 15.0 %, PC2 11.8 %
G · K-means	k-Means (k = 6), GroupBy + Unpivot + Heatmap, cluster Scatter Plot, chord Python View	Chemistry clusters compared with family names
H · Circular visuals	Sunburst Chart, radar Python View	Group → family hierarchy
I · Machine learning	Number to String, Row Splitter, Column Filter, Decision Tree, Normalizer + K Nearest Neighbor, Random Forest (Regression), Scorers	See Table 13

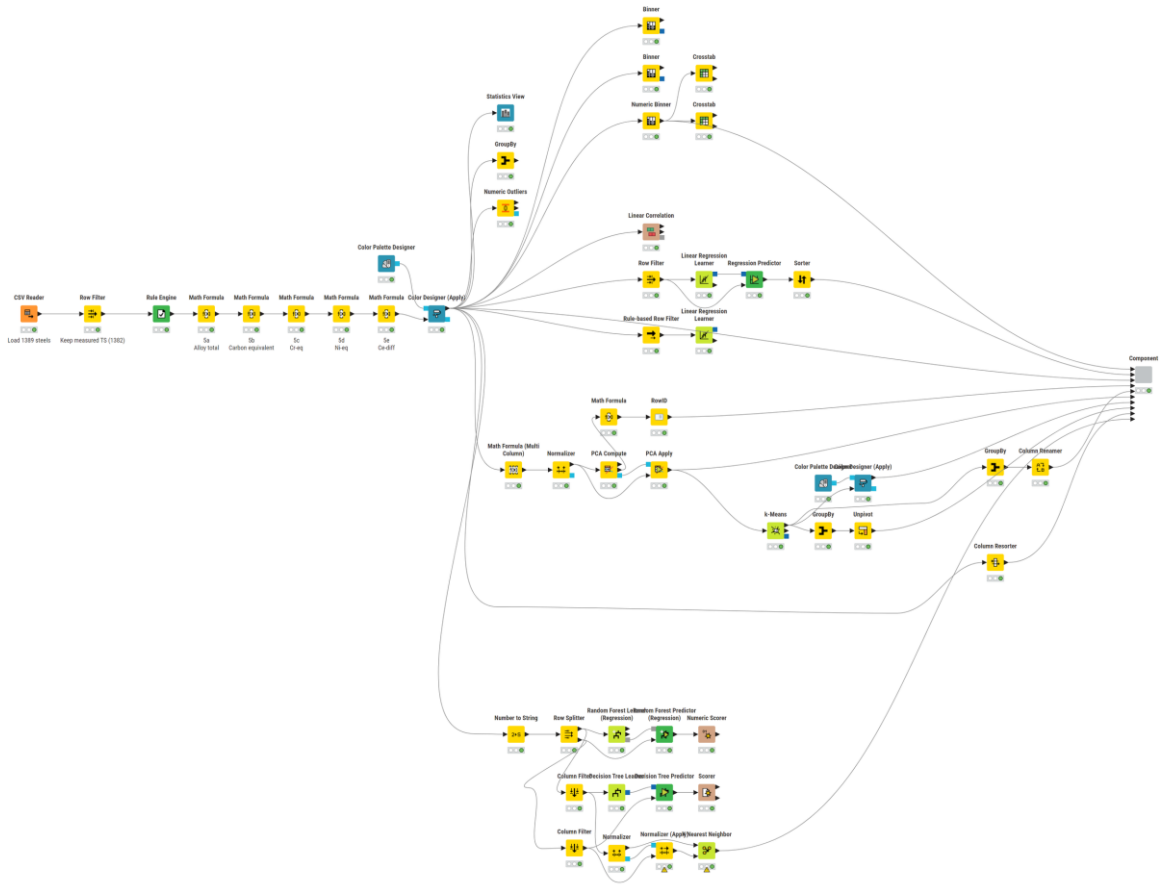


Figure 21: *The complete KNIME workflow: the read-clean-enrich trunk (left) feeds the statistics, binning, chi-square, regression, PCA and k-means branches and the machine-learning section (bottom); the view nodes are collected in the dashboard Component (right).*

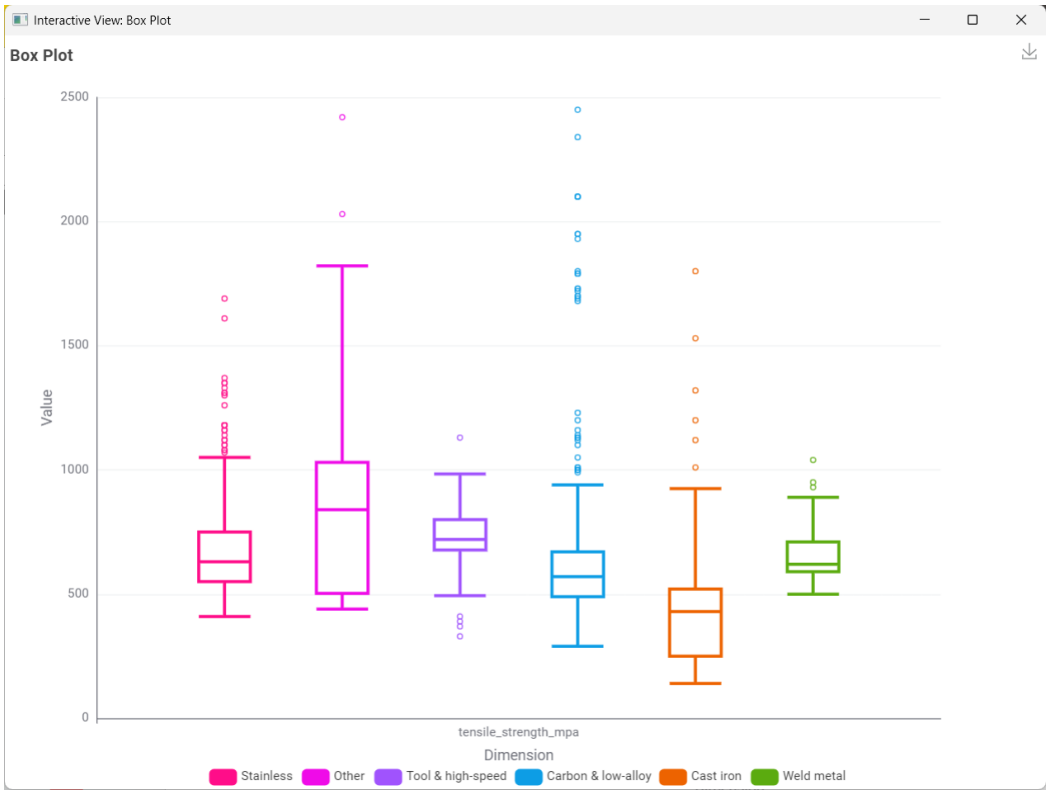


Figure 22: *KNIME Box Plot view: tensile strength by family group.*

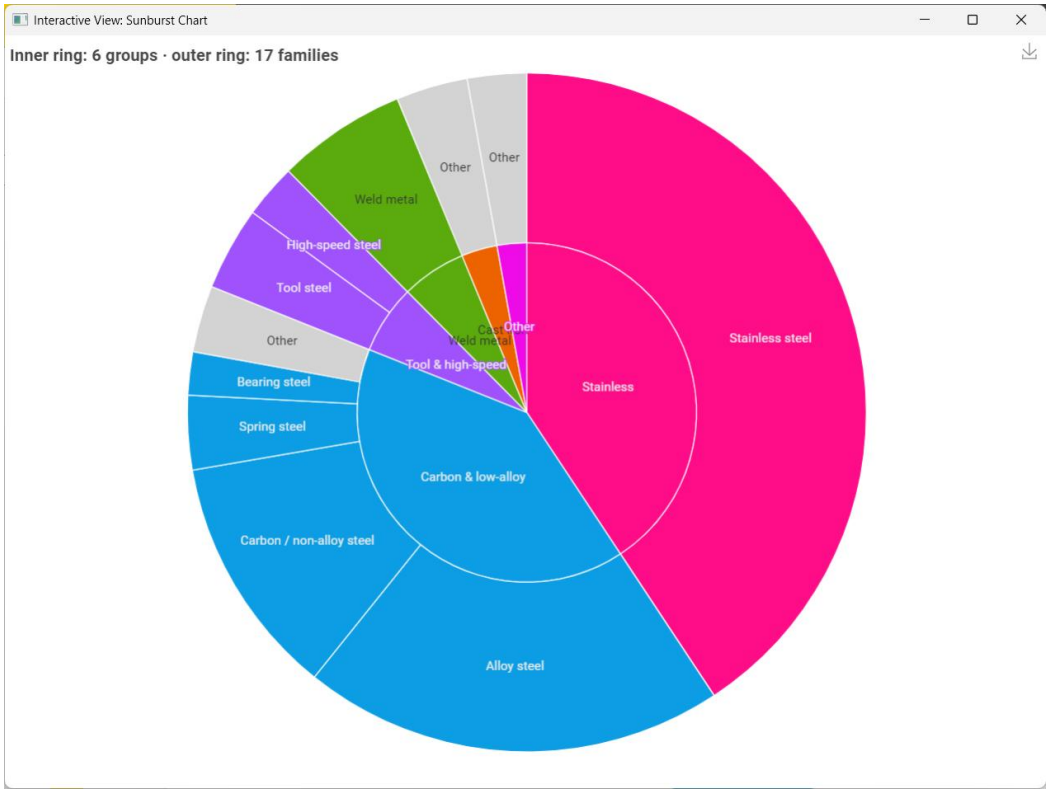


Figure 23: *KNIME Sunburst view: family groups (inner ring) and families (outer ring).*

7.1 Clustering

k-means was run on the same 20 log-transformed, standardised composition columns used for PCA, with six clusters to match the six family groups. A silhouette check in Python on the same data rises slowly from 0.23 (k = 2) to 0.34 (k = 12) with a local peak of 0.30 at k = 6, supporting the choice. The cluster heatmap (median z-score per element) shows distinct fingerprints: a large lean cluster of plain steels, Cr–Ni austenitic stainless, Cr-only stainless, W–V–Mo–Co tool steels and C–Si cast irons. The chord diagram from family group to cluster shows that plain steels form one chemistry under many names while stainless steels split into several chemistries under one name.

7.2 Machine learning

All models use composition, the engineered indices, family and product form as inputs. The split is group-safe: grades with an identical chemistry never appear in both training and test data (1,117 training and 265 test grades), so the models cannot score by memorising a twin. Tensile strength is excluded from the classifiers' inputs because the strength class is defined from it, and the strength class is excluded from the random forest for the same reason. For k-NN the inputs are Min-Max scaled, fitted on the training set only and applied to the test set.

Table 13: *Machine-learning results in KNIME*

Model	Task	Test result (265 grades)	Baseline
k-nearest neighbours (k = 5, distance-weighted)	Strength class	Accuracy 0.842, Cohen's κ ≈ 0.70	Always "High": accuracy 0.502
Random forest (100 trees)	Tensile strength	R^2 0.731, MAE 66.5 MPa, RMSE 95.8 MPa, MAPE 11.6 %	Predict the mean: MAE 134.4 MPa

Model	Task	Test result (265 grades)	Baseline
Decision tree (Gini, MDL pruning)	Strength class	Read from the workflow's Scorer node after the train/test correction below	Accuracy 0.502

All k-NN errors fall between the medium and high classes near the 600 MPa boundary, where heat treatment rather than chemistry decides; no low-strength grade was classed as high. The random forest halves the error of the mean predictor.

Lessons from building the models. (1) The first decision tree scored 100 % because tensile strength was still among its inputs; a perfect score on unseen data revealed target leakage, fixed with a Column Filter. (2) A single Column Filter initially fed both the learner and the predictor, so the model was tested on its own training rows; separate filters for the training and test outputs fixed it. (3) Without scaling, large-range columns such as Ni_eq (0–110) dominate the k-NN distance; scaling raised accuracy from 0.815 to 0.845 in a Python check. (4) The Normalizer warned that some test values fall outside the training range, which is expected when the test data is genuinely unseen.

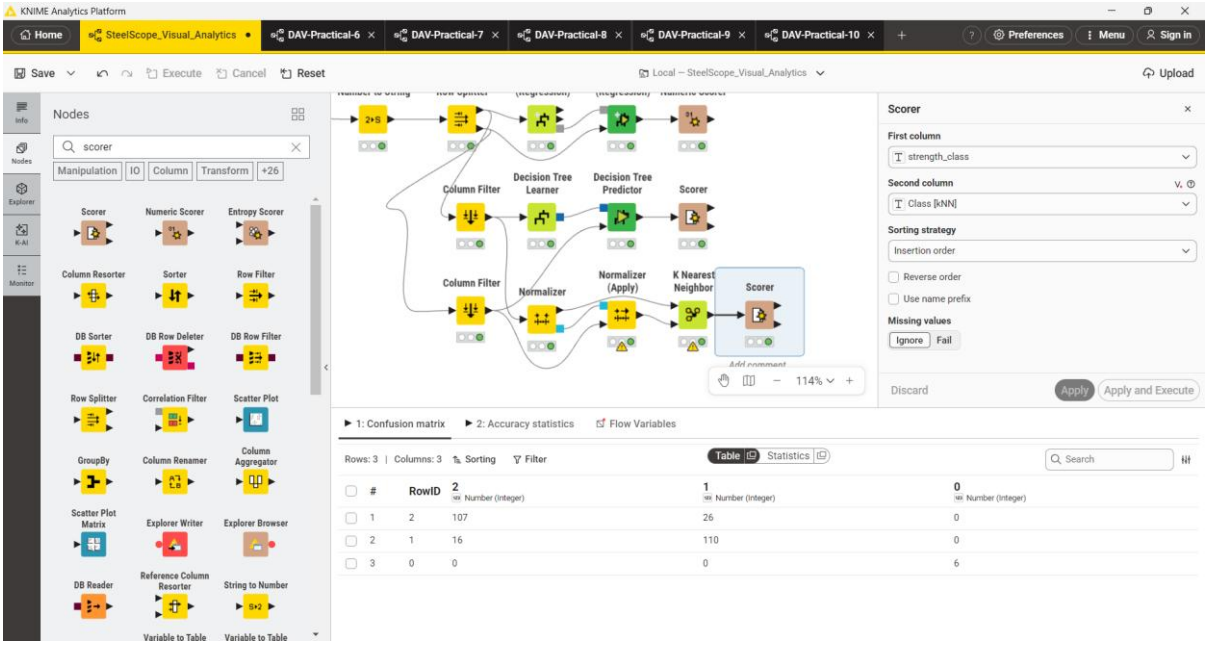


Figure 24: *KNIME machine-learning section: separate training and test branches, and the k-NN confusion matrix.*

8. Tableau dashboards

The Tableau workbooks (tableau/workbooks) connect to the CSV files exported by notebook 03 (tableau/data). Charts that Tableau cannot draw natively, such as chord, radar and waterfall charts, are supplied as ready-made x/y shapes computed in Python, so Tableau only connects the points.

Table 14: *Tableau workbooks*

Workbook	Sheets	Data
DAV_D1	Six sheets and a dashboard	steels_wide.csv
DAV_D2	Attribute vs Cramér's V, PCA analysis, regression line, term vs beta (coefficients), treemap, and a dashboard	steels_wide.csv and the chi-square, PCA and regression tables
DAV_D3	Box plot MP, Box plot TS, MP vs Al, TS in Alloys, TS vs MP, two further sheets and a dashboard	Synthetic_Alloys_Tableau.csv

9. Power BI report

The Power BI report (powerbi/SteelScope.pbip) is built on a star schema generated from the cleaned data: a grades fact table (1,389 rows) with composition (13,376 grade–element rows), families (17) and elements (30) as related tables. Power Query loads the tables through a DataFolder parameter, DAX measures compute the statistics (for example grades 1,389, median strength 610 MPa, middle 50 % 510–720 MPa, median CE 2.53), and a theme applies the project's family-group colours.

Table 15: *Pages of the Power BI report*

Page	Question	Visuals
Overview	What is in the data and how strong is it?	Slicers, KPI cards, strength histogram, family treemap, quartiles by group
Families & classes	Does family decide strength?	100 % stacked bars, class matrix heatmap, decomposition tree
Carbon is the dial	What makes a plain steel stronger?	Carbon scatter with trend line, pooled scatter, weldability bands
Chemistry map	Which steels are chemically alike?	PCA scatter, element heatmap matrix
Grade inspector	What is this one grade?	Drillthrough page: cards, composition bars and table
Target finder	Which grades meet my requirement?	Range slicers, matches per family, candidate table

10. Web application: SteelScope

The notebooks tell the story once; the web application lets any user ask their own question of the same data in a browser. It is built with Next.js, React, TypeScript, Tailwind CSS, shadcn/ui and Recharts, pre-renders 2,787 pages and can be deployed on Vercel. A Python script prepares its data from the cleaned dataset with the same PCA as notebook 03. The application does not predict properties; machine learning stays in KNIME.

Table 16: *Pages of the web application*

Page	What it does	Why it is useful
Explorer	Filters by family, form, strength and name; stat cards, stacked histogram, sunburst, box plots, treemap, chord diagram, sortable table	Shows the whole dataset at a glance and recomputes every chart for the user's filter
Grade inspector	Percentile of a grade within its family and overall, composition radar against the family median, composition bars, CE waterfall with weldability verdict, five closest grades	Puts a single strength number in context in seconds
Substitute finder	Ten grades with the closest chemistry, distance, strength difference, PCA map	Shortlists substitutes when a grade is unavailable or named in a foreign standard
Target designer	Strength and liquidus windows, product form, placeholder switch; candidates per family	Turns a requirement into a list of candidate grades
About the data	Cleaning log, methods and limits	Makes every number traceable

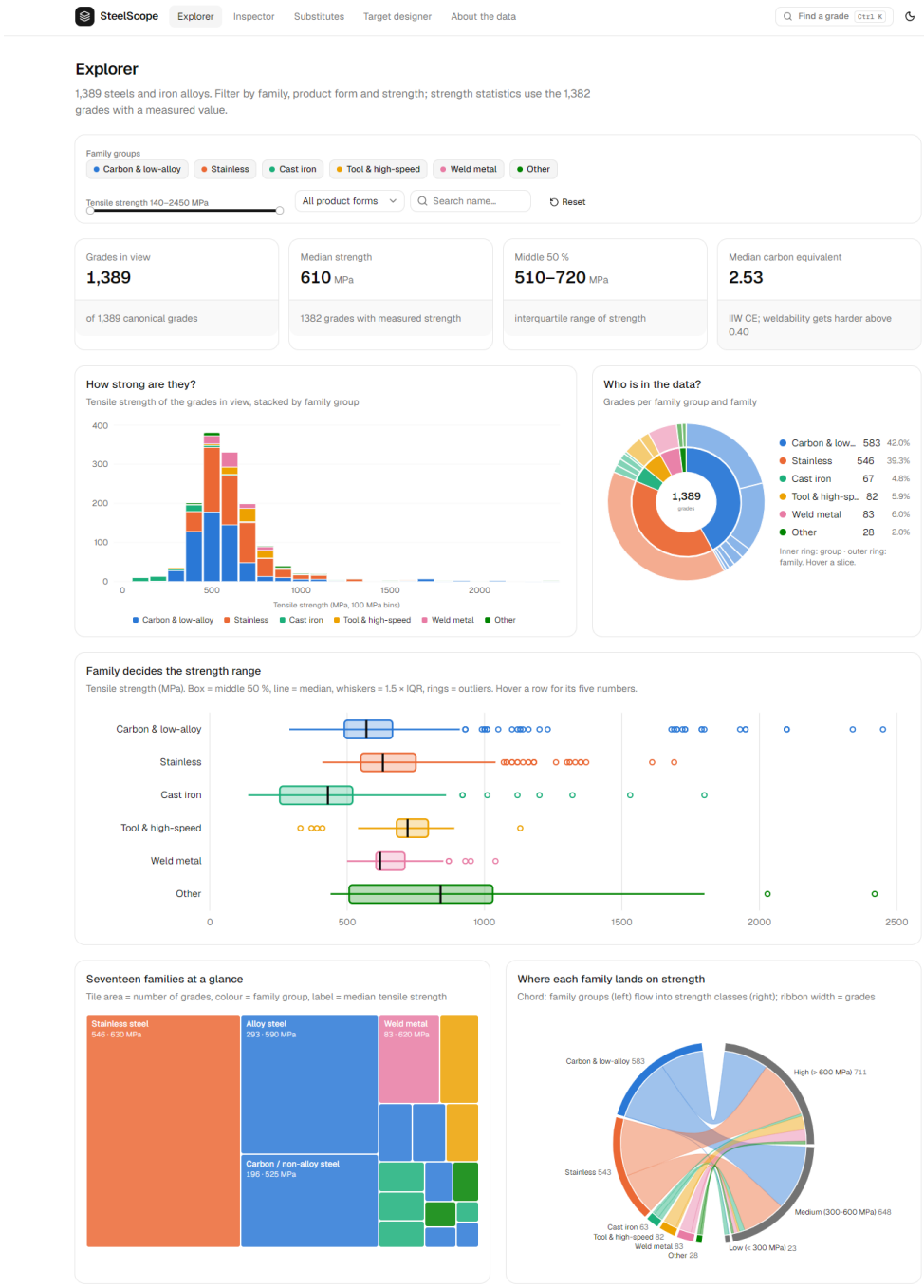


Figure 25: *SteelScope Explorer: filters, statistics, histogram, sunburst and box plots.*

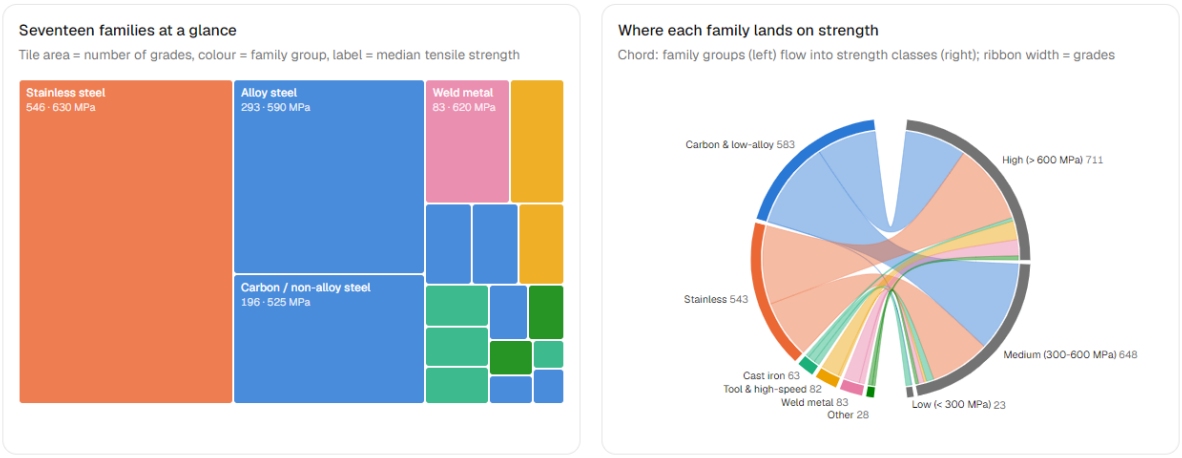


Figure 26: *Explorer: treemap of the 17 families and chord diagram from family group to strength class.*

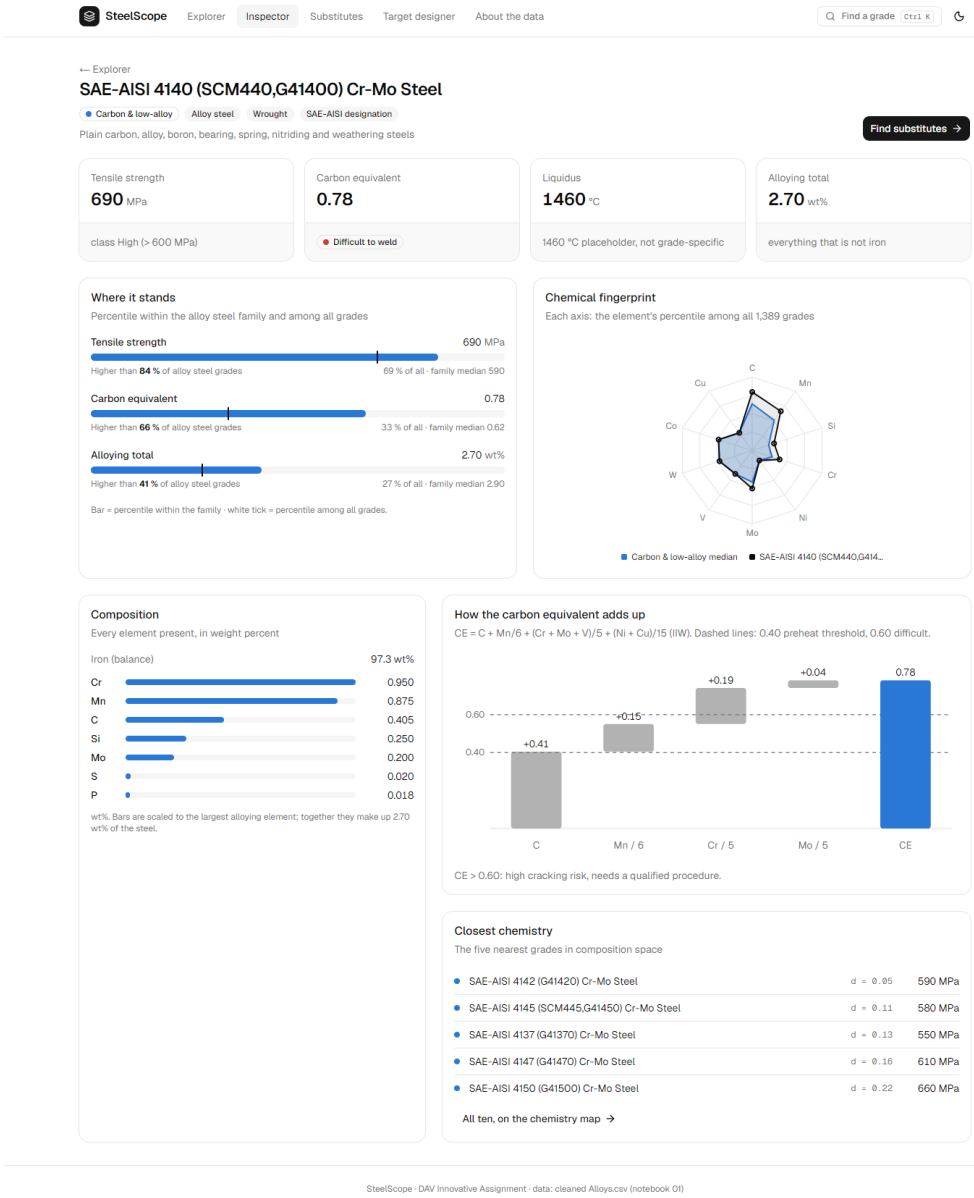


Figure 27: *Grade inspector for SAE-AISI 4140: 690 MPa, stronger than 84 % of alloy steels, CE 0.78 (difficult to weld).*

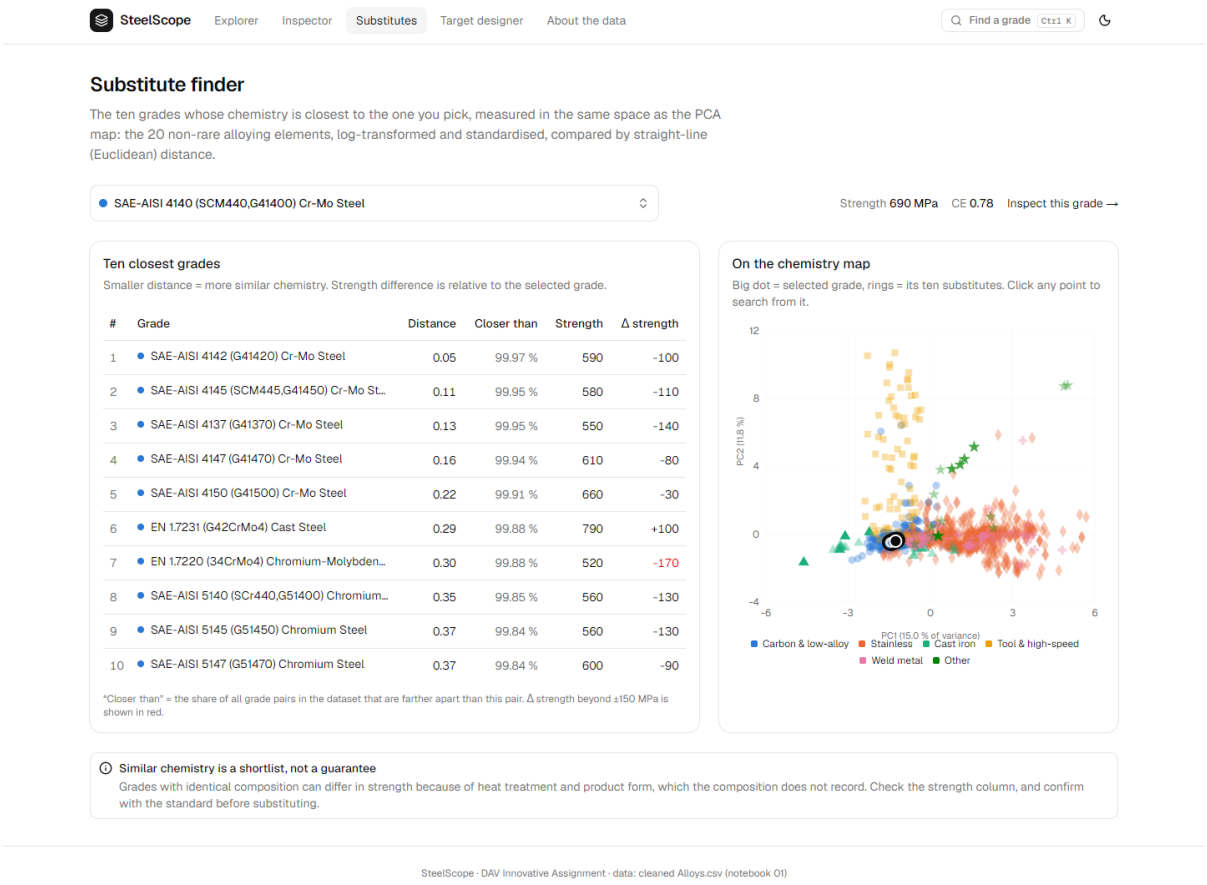


Figure 28: *Substitute finder for AISI 4140: 4142, 4145, 4137, 4147 and 4150 first, then European equivalents.*

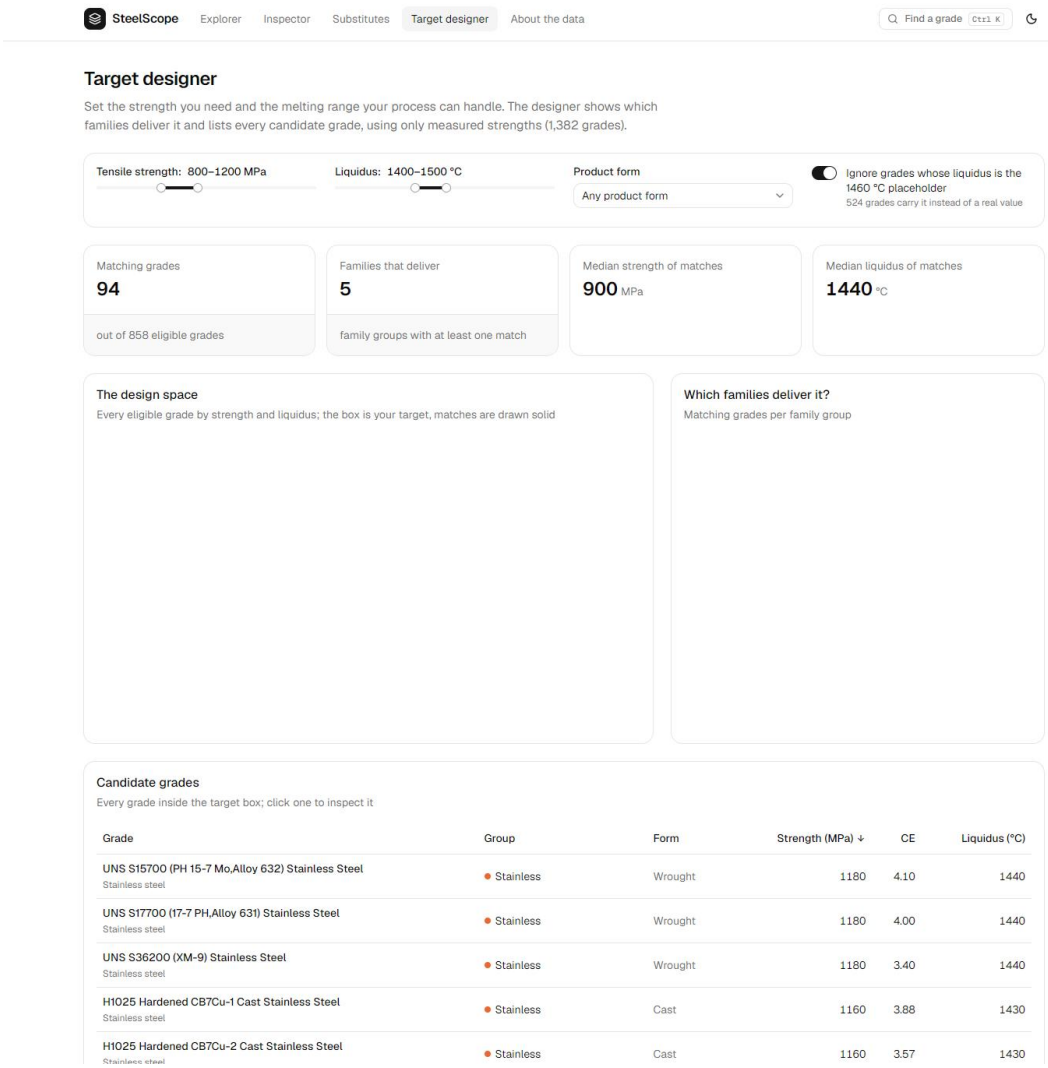


Figure 29: *Target designer: 800–1200 MPa and 1400–1500 °C give 94 candidate grades, 68 of them stainless.*

11. Key findings

1. **Clean before you count.** 2,672 rows were really 1,389 steels; the fake 652 MPa values hid inside duplicate listings.
2. **Think in medians.** Strength is right-skewed; the typical grade has 610 MPa, and the middle half lies between 510 and 720 MPa.
3. **Family first, chemistry second.** The family sets the strength range ($\chi^2 = 548.7$, Cramér's V = 0.45); elements fine-tune it.
4. **Carbon is the plain steel's dial:** about +43 MPa per 0.1 % carbon ($R^2 = 0.51$), but only within plain steels (Simpson's paradox).
5. **Names are not chemistry.** On the PCA map and in the k-means clusters, many plain-steel names share one chemistry while stainless splits into several.
6. **Strength is not bought by the kilogram.** 17-4 PH beats nickel-rich 304 and 316 on strength with less nickel.
7. **Composition tells most of the story; processing writes the rest.** A random forest explains 73 % of the strength variance on unseen chemistries; the remainder is heat treatment.

12. Limitations and future work

- Heat treatment and product condition are not recorded, so identical chemistries can differ in strength; this limits every composition-based model.
- Seven grades lack a measured strength and 524 liquidus values are a placeholder; both are flagged and excluded where they would bias results.
- The low-strength class has only 23 grades, so class-level results for it are uncertain; permutation tests backed the chi-square results where expected counts were small.
- Future work: add processing data (temper, ageing condition) to separate processing variants, add further properties (yield strength, hardness, elongation) and cost, and extend the substitute finder with property constraints.

13. Conclusion

The project shows that an honest cleaning step changes the conclusions of a dataset: the raw table double-counted most grades and contained hundreds of invented strength values. On the cleaned data, a combination of descriptive statistics, binning, chi-square tests, regression and PCA establishes that the family decides the strength range, that carbon controls strength within plain steels, and that grade names group chemistries imperfectly. The same results are reproduced independently in KNIME, where the machine-learning models reach a test accuracy of 0.84 (k-NN) and an R^2 of 0.73 (random forest) on chemistries they never saw. Tableau and Power BI dashboards and the SteelScope web application make the analysis usable for engineers and buyers who need to explore grades, find substitutes and select materials for a target.

References

[1] International Institute of Welding, carbon-equivalent formula CE = C + Mn/6 + (Cr + Mo + V)/5 + (Ni + Cu)/15.

[2] A. L. Schaeffler, constitution diagram for stainless steel weld metal, Metal Progress, 1949.

[3] W. D. Callister and D. G. Rethwisch, Materials Science and Engineering: An Introduction, Wiley.

[4] F. Pedregosa et al., Scikit-learn: Machine Learning in Python, Journal of Machine Learning Research 12, 2011.

[5] KNIME Analytics Platform documentation, KNIME AG, knime.com.

[6] H. Cramér, Mathematical Methods of Statistics, Princeton University Press, 1946 (Cramér's V).

[7] Microsoft Power BI documentation; Tableau documentation; Next.js and shadcn/ui documentation.

Appendix A: Repository structure

Table 17: *Repository folders*

Folder	Contents
data/raw, data/processed	Original Alloys.csv (never edited) and the cleaned tables written by notebook 01
notebooks/	01_cleaning, 02_eda, 03_visual_analytics
reports/	39 figures and the cleaning log
knime/	Guides, the .knwf workflow, its data and the Python View scripts
tableau/	Guide, workbooks DAV_D1–D3 and their data
powerbi/	Power BI project (.pbip), star-schema data, Power Query, DAX measures, theme and guide
app/	The SteelScope web application and its data script
docs/	Guide index, project journey (decision log), reusable prompt